

Tide retrospective: base-coefficient recovery (unequal-base programme, stages B1–B2)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

1 Setting

The weighted-jet recovery theorem compares two jets over the *same* base ax^{2k} . The germbij Theorem 3.1 setting has the quadratic coefficient unknown: λ is itself recovered from the data, first, from the t -rate of the second moment. The scoping consult for the post-programme work split the remainder into a small programme B' (unequal bases, five stages, this tide is B1–B2) and a separate programme C (the smooth-germ transfer, which needs an asymptotically *stable* recovery theorem, since Taylor replacement never produces exactly equal data). The consult's structural advice: do not extend the pairwise-difference machinery to unequal bases — at $a_1 \neq a_2$ the difference has a q^0 term, so the q^{-r} argument is at the wrong scale by design; recover the base first from the leading order, then apply the existing theorem verbatim.

2 The thing formalised

Six declarations in `Laplace/OneD/BaseRecovery.lean`, sorry-free with zero warnings (gate verified via the import line and the fresh `.olean`, per the incident rule). The second moment of the jet Gibbs measure converges to the reference ratio $M_2(a)$ as $q \rightarrow 0^+$; the ratio has the closed form

$$M_2(a) = (1/a)^{1/k} \frac{\Gamma(3/(2k))}{\Gamma(1/(2k))},$$

strictly decreasing in a for every $k \geq 1$ (for $k = 1$ this is the familiar $1/(2a) = 1/\lambda$); so eventually-equal second moments force equal bases¹. A pleasant discovery en route: the profile envelope evaluated at $y = 0$ is a itself, so $0 < a$ is a consequence of `HasPositiveJetProfile`, not an extra hypothesis (`HasPositiveJetProfile.base_pos`).

3 Proof notes

Everything reduces to existing machinery: the base-point DCT limit from the recovery tide (numerator and denominator separately, then `Tendsto.div`), the $t = a(2k)!$ bridge to the monomial-potential Gamma API (as in the variance bridge), and `rpow` base-monotonicity for injectivity. The Gamma-ratio cancellation is cleanest as `mul.right.cancel₀` around a three-step `ring` calc — `field.simp` on hypotheses of this shape produces unpredictable normal forms. Two small reusable

¹`base_recovery`; the limit-based variant `base_recovery_of_tendsto` is the designed interface for programme C.

facts: a def under a `Tendsto` binder unfolds by `.congr fun q => rfl` (plain `rw` cannot); and exponent normalisations inside integrands belong at the integral level (`integral_congr_ae + norm_num`), never as lambda-level rewrites.

4 The deliberation and check

The scoping consult is archived in the tide log with the votes; its stage list makes B3 (variable-base recovery) a verbatim composition of this tide with `polynomialJet_recovery`, and B4–B5 the $k = 1/\lambda$ -facing packaging. The numerical check (executed first) matched the Gamma form to quadrature at eight decimals for $k \in \{1, 2\}$, watched the jet moment converge to it, and confirmed strict monotonicity on a grid.

5 What the next tide inherits

Stages B3–B5 in one packaging tide: compose base recovery with the comparison theorem into `polynomialJet_recovery_variableBase`, specialise to $k = 1$ with $\lambda/2$ coefficients, and provide the t -facing corollaries ($t\mathbb{E}_t[x^2] \rightarrow 1/\lambda$). After that, programme C — the asymptotically stable recovery theorem and the smooth-loss localisation — is the last genuinely analytic gap between the formalised polynomial core and the note’s Theorem 3.1.